

of Proposed Features Features

Date	15 March 2026
Team ID	XXXXXXX
Project Name	Rising Waters – Flood Prediction Using Machine Learning
Maximum Marks	1 Mark

of Proposed Features: Features:

The Flood Prediction using Machine Learning project was developed to provide users with an accurate and user-friendly platform for predicting flood risk based on environmental parameters. All major features proposed during the planning phase were successfully implemented, tested, and demonstrated. The application integrates a trained Machine Learning model with a Flask web interface, allowing users to enter input values and receive instant flood risk predictions. Each implemented feature was verified during system testing to ensure reliability, accuracy, and ease of use.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	User Input Interface	Developed a Flask-based web page to accept rainfall, humidity, river level, temperature, and other environmental parameters from the user.	Implemented	Yes	User-friendly interface with input validation.
2	Machine Learning Flood Prediction	Integrated the trained Machine Learning model to predict flood occurrence based on the entered parameters.	Implemented	Yes	Generates accurate predictions using the trained model.
3	Risk Level Display	Displays prediction results along with clear flood risk indicators	Implemented	Yes	Provides easy-to-understand

		such as Low Risk or High Risk.			output for users.
4	Data Preprocessing	Automatically preprocesses and formats user input before passing it to the prediction model.	Implemented	Yes	Ensures consistent and reliable predictions.
5	Performance Evaluation	Evaluated the Machine Learning model using accuracy, precision, recall, F1-score, and confusion matrix.	Implemented	Yes	.
6	Local Flask Deployment	Successfully deployed the application on a local Flask server and verified end-to-end functionality.	Implemented	Yes	Application runs successfully through a web browser.

Implementation Summary: Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%